

AI-DRIVEN PLACEMENT PREDICTION AND INSIGHTS

¹DR MANJU PAWAR, ²SUMEDH CHOUDHARI, ³SOHAM PHUTANE, ⁴ANUSHKA JADHAV,
⁵TEJAS BHOR

¹Department of Artificial Intelligence and Data Science, Zeal College of Engineering and Research, Maharashtra, India
E-mail: ¹manju.pawar@zealeducation.com, ²sumedhchoudhari02@gmail.com, ³phutanesoham@gmail.com,
⁴anushkajadhav966@gmail.com, ⁵tejasbhor2003@gmail.com

Abstract - A comprehensive, data-driven platform called "AI-Driven Placement Prediction and Insights" was created to help academic institutions and students better understand and improve placement preparedness. The system combines a number of contemporary technologies to provide precise placement forecasts and useful information. Flutter Web was used in the development of the user interface, making it responsive and fluid.

Primarily, the platform utilizes a Random Forest algorithm-based machine learning model that was trained on a variety of student characteristics, such as academic performance (10th, 12th, and graduation grades), quiz scores, technical scores (out of 20), logical reasoning, verbal ability, quantitative aptitude, the number of projects and internships completed, and mastery of critical skills like SQL, DSA, Java, Python, C++, ML, AI, GenAI, Tableau, ReactJS, NodeJS, and more. Real-time predictions based on the most recent student inputs are ensured by exposing the model through a Flask API that is implemented on Render.

The creation of comprehensive Power BI dashboards is made possible by the system's safe management and storage of all data. These dashboards help administrators and students identify placement trends, skill gaps, and overall performance data by providing relevant visuals.

To ensure scalability and accessibility, the complete system is integrated into the Flutter Web application, and Firebase Hosting is used for the final release. This project demonstrates the usefulness of AI and ML in the educational field as well as the ability of full-stack integration to provide insightful, engaging, and significant solutions.

Keywords - Placement Analysis, Placement Prediction, Machine Learning, Data Visualization, Data-Driven

I. INTRODUCTION

In today's competitive job market, student placements play a critical role in shaping the future of graduates and determining the reputation of educational institutions. However, predicting placement outcomes and identifying the factors influencing employability remain challenging tasks due to the dynamic nature of industry demands and student profiles.

This project, Placement Analysis and Prediction Using Machine Learning, addresses these challenges by leveraging the power of data science and predictive modeling.

By analyzing historical placement data and current student attributes, the system provides actionable insights into placement trends and predicts the likelihood of students securing job offers. The project aims to empower students and academic administrators to make informed decisions by detecting talent shortages and aligning their efforts with industry requirements.

The system incorporates machine learning algorithms to analyze features such as programming skills, academic performance, extracurricular involvement, and recruitment patterns. Through visualizations and predictive analytics, the platform offers a user-friendly interface that highlights trends and provides

recommendations for improvement. Additionally, the tool ensures scalability, data security, and ease of access, making it suitable for diverse educational institutions.

This project bridges the gap between academic preparation and industry expectations, enabling institutions to enhance their placement strategies and helping students unlock their career potential.

II. HELPFUL LITERATURE REVIEW

The development of predictive systems for student placements has seen significant advancements in recent years, primarily due to the application of machine learning (ML) techniques. Researchers like Milind Ruparel et al. (2024) and Advait Joshi et al. (2024) have demonstrated the effectiveness of supervised learning algorithms such as Logistic Regression, Random Forest, and Decision Trees in predicting placement probabilities. These algorithms leverage features like academic performance, technical skills, and extracurricular activities to enhance prediction accuracy.

Several studies emphasize the importance of feature selection and data preprocessing in improving model performance. For instance, Rushikesh Joshi et al. (2024) highlighted how including factors like internships and project work can significantly refine predictions. Similarly, Vinod Agrawal et al. (2023)

reviewed various ML algorithms, finding Decision Trees and Random Forests particularly effective in placement prediction.

Moreover, modern predictive systems aim to go beyond mere prediction by offering actionable insights. Priyanka Shahane (2022) and Varsha Harihar (2020) underscored the importance of integrating visualization tools and analytical features to identify skill gaps and recommend interventions. These systems are not limited to predictions but also provide strategic guidance to institutions for enhancing placement outcomes.

The literature collectively underscores the potential of ML-driven predictive tools in transforming traditional placement processes into data-driven frameworks. This lays a strong foundation for the proposed "Placement Analysis and Prediction" project, which aims to implement a robust and accurate system tailored for real-world educational applications.

III. METHODOLOGY

The implementation of the Placement Prediction and Analysis project can be divided into several key phases, each encompassing specific tasks and objectives. These phases guide the systematic development and deployment of the project:

1. Data Collection

- Sources: Historical placement data from the college's database, including:
- Academic scores (e.g., GPA).
- Technical skills and certifications.
- Internship and project details.
- Extracurricular activities.
- Placement outcomes.
- Format: Data is collected in a structured format (e.g., CSV files).

2. Data Preprocessing

- **Data Cleaning:**
- Handling missing values (e.g., using mean/mode imputation).
- Removing or correcting outliers.

- **Data Transformation:**

Encoding categorical data (e.g., programming languages known) into numerical form.

- **Feature Engineering:**

Recognizing and developing novel traits that might affect placement results.

3. Model Development

- **Algorithm Selection:**

The selection of algorithms for classification problems is dependent on factors including Neural

Networks, Random Forests, Decision Trees, and Logistic Regression.

- **Model Training:**

- Splitting the data into training (80%) and testing (20%) sets.
- Using cross-validation to ensure robustness and avoid overfitting.

4. Model Evaluation

- **Metrics:** Evaluating the models using metrics like:
- Accuracy
- Precision
- Recall
- F1-score
- **Comparison:** Comparing the performance of various models to select the best-performing one for deployment.

5. Insights and Recommendations:

Branch performance insights

- Evaluates effectiveness of individual branches.
- Identifies top performers and areas for improvement.

Yearly placement analysis

- Examines placement trends and statistics across different years.
- Reveals long-term patterns and growth in placements.

Company insights: leading placement volume

- Identifies companies recruiting the most students (placement volume).

Uncovering trends by location, department, And role:

- Analyzes placement patterns based on geographic location, academic department, and specific job roles.
- Helps understand skill demand and curriculum needs.

6. Visualization and Reporting

Visualization Tools: Creating dashboards to display placement probabilities, influencing factors, and trends using tools like PowerBI or Python visualization libraries.

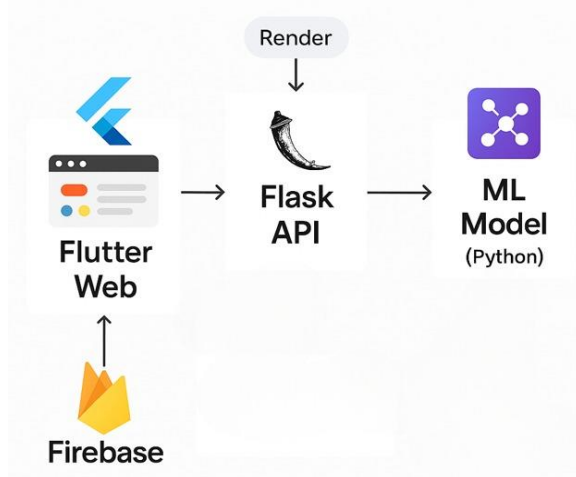
Reports: Generating reports summarizing placement insights and predictions.

7. Deployment

- Platform: Deploying the system as a web or mobile application for ease of access.
- Tools: Using platforms like Flask, Flutter for web deployment, with cloud support (Render).

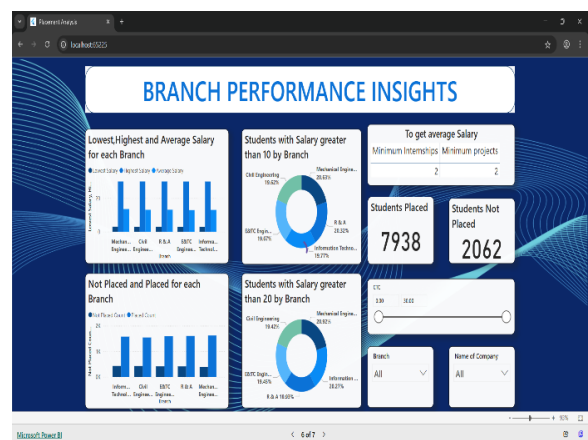
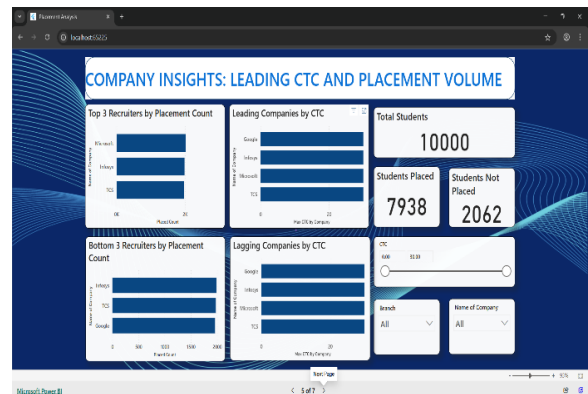
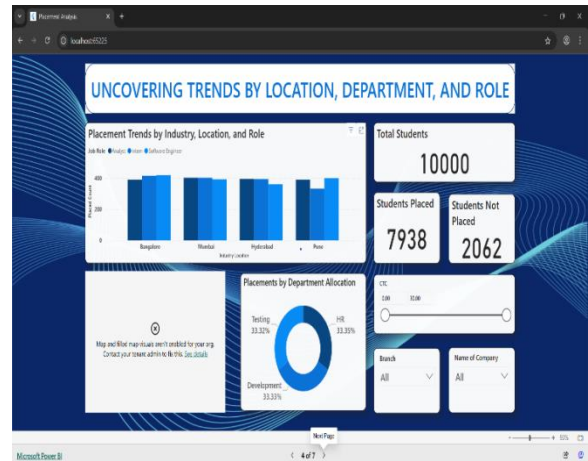
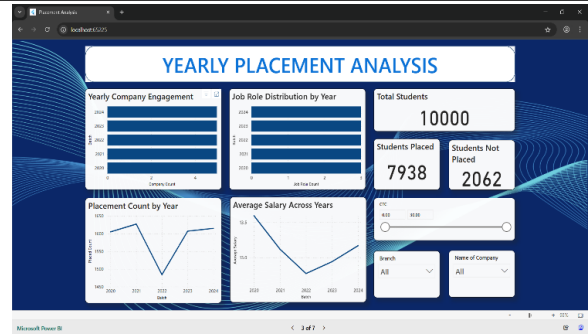
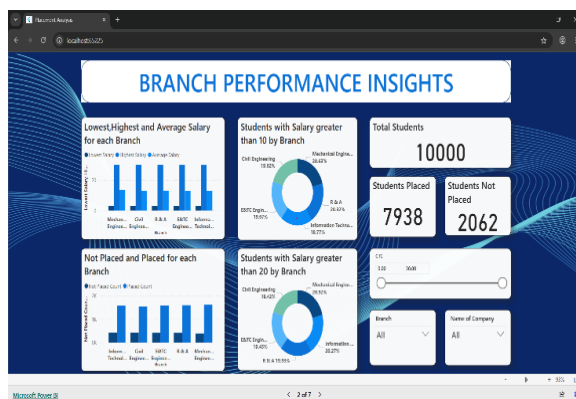
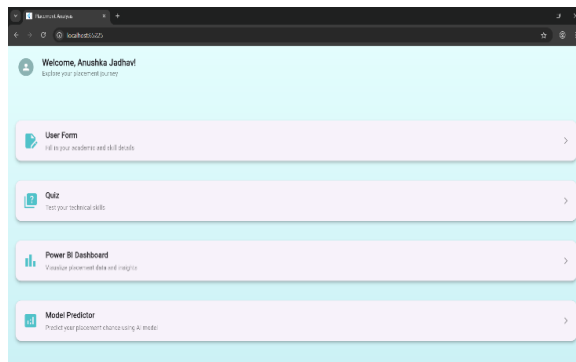
8. Testing and Validation

- User Testing: Collecting feedback from students and administrators to refine the system.
- Performance Validation: Ensuring the model performs well on unseen data and aligns with real-world requirements.



IV. RESULTS

Random Forest achieved a perfect accuracy of 100%, with precision, recall, and F1-score all equal to 1.00 for both classes. While this may indicate strong predictive capability, such perfect results could also suggest possible overfitting, especially in real-world applications.



User Form

Full Name: Anushka Jadhav
 Email: anushka@gmail.com
 Phone Number: 1234567890
 10th Marks (%): 85
 12th Marks (%): 95
 Number of Internships: 3
 Number of Projects: 5
 Languages (English, Hindi, etc.): English, Hindi, Marathi
 Technical Skills: Python, Java, WebDev, ReactJS
 COPE (0-10): 8

Placement Prediction

Test Results

Tech Metrics

Metric	Score (0-10)
Overall Marks	85.00%
Technical Skills	85.00%
Communication Skills	85.00%

Placement Prediction Probability: 58.00%

Recommendations

- Improve your technical skills by practicing coding challenges.
- Consider a part-time internship to gain practical experience.
- Develop a strong portfolio of projects to showcase your skills.
- Focus on improving your communication and presentation skills.
- Consider a short-term course or workshop to learn new technologies.
- Network with professionals in your field of interest.
- Keep your resume and LinkedIn profile up-to-date.

V. CONCLUSION AND FUTUREWORK

The dataset's class imbalance issue is demonstrated by the consistently declining macro F1-scores for all model but Random Forest. Overall, the study shows that Random Forest is the most robust and complete model for this prediction task. The Random Forest classifier outperformed the other machine learning techniques with perfect results in accuracy, precision, recall, and F1-score. This demonstrates its high predictive capacity, but it also raises concerns about potential overfitting, which requires further investigation. The relatively good performance of other models, such as Logistic Regression, XGBoost, and SVM, but their difficulties with class imbalance brought attention to the importance of dataset balancing and model tuning.

The study also highlights how important it is to assess and display model results using intuitive dashboards and analytics. In addition to helping administrators make data-driven strategic choices, these visual aids assist students in identifying their areas of strength and progress. Project experience, internships, programming skills, and academic accomplishment were shown to be significant indications that closely matched the demands of the real industry.

All things considered, this study demonstrates how educational institutions may apply these findings to improve placement outcomes and validates that machine learning is a valid technique for predicting placement prevalence. Because prediction, analysis, and recommendation are all integrated into a single,

coherent platform, this method is both practical and expandable. By closing the gap between academic success and industry readiness, it creates the foundation for more astute, data-driven institutional planning and career advising.

FUTURE WORK

While the current model shows promising results, several avenues exist for further improvement:

1. **Addressing Class Imbalance:** Future iterations should implement advanced techniques such as SMOTE, ADASYN, or cost-sensitive learning to mitigate class imbalance and improve performance on minority classes.
2. **Incorporating Real-Time Data:** Integrating dynamic or real-time student performance data (e.g., ongoing project performance, recent certifications) could enhance prediction accuracy.
3. **Model Interpretability:** Introducing explainable AI (XAI) methods can help users understand the rationale behind predictions, increasing transparency and trust in the system.
4. **Deployment at Scale:** Future development can focus on making the system scalable across multiple institutions, integrating APIs, cloud-based storage, and robust security mechanisms.
5. **Personalized Recommendations:** Enhancing the system to provide personalized roadmaps for individual students based on their unique strengths and weaknesses can significantly boost their placement chances.

REFERENCES

- [1] Joshi, A. Hutke, A. Thete, D. Aware, S. Rawale, and N. A. Kandalkar, "College placement prediction and analysis using data science," *International Research Journal of Modernization in Engineering Technology and Science*, vol. 6, no. 3, pp. 4336–4340, Mar. 2024. [Online]. Available: [www.irjmets.com/contentReference\[oaicite:0\]{index=0}](http://www.irjmets.com/contentReference[oaicite:0]{index=0}).
- [2] M. Ruparel and P. Swaminarayan, "Student placement prediction using various machine learning techniques," *International Journal of Intelligent Systems and Applications in Engineering*, vol. 12, no. 3, pp. 2107–2113, Mar. 2024. [Online]. Available: [www.ijisae.org/contentReference\[oaicite:1\]{index=1}](http://www.ijisae.org/contentReference[oaicite:1]{index=1}).
- [3] K. Srinivas, N. S. Yadav, P. A. Somashekar, and S. K. R., "Students placement prediction using machine learning," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, vol. 8, no. 5, pp. 2771–2774, May 2020. [Online]. Available: [www.ijraset.com/contentReference\[oaicite:2\]{index=2}](http://www.ijraset.com/contentReference[oaicite:2]{index=2}).
- [4] V. S. Agrawal and S. Shelke Kadam, "Review on predictive analysis of placement of students using machine learning algorithms," in *ICAMIDA 2022*, Aurangabad, MH, India: MGM University, 2023, pp. 23–30. doi: 10.2991/978-94-6463-136-4_4(3).
- [5] P. Shahane, "Campus Placements Prediction & Analysis using Machine Learning," in *2022 International Conference on Emerging Smart Computing and Informatics (ESCI)*, Pune, India, Mar. 2022, pp. 1–5, doi: 10.1109/ESCI53509.2022.9758214(1).
- [6] V. K. Harihar and D. G. Bhalke, "Student Placement Prediction System using Machine Learning," *SAMRIDDHI: A Journal of Physical Sciences, Engineering and Technology*, vol. 12, no. SI-2, pp. 85–91, 2020, doi: 10.18090/samriddhi.v12iS2.17(7).

★ ★ ★